

AI/ML Engineer

Web Portfolio : <https://lpsangg.github.io/lpsang/>
Linked: <https://www.linkedin.com/in/lpsang/>
Github : <https://github.com/lpsangg>

SUMMARY

AI/ML Engineer with hands-on experience in building and deploying machine learning models and AI-driven applications. Skilled in Python, Machine Learning, Deep Learning, data preprocessing, and developing end-to-end AI pipelines using Flask/FastAPI. Strong foundation in algorithms, data structures, SQL, and computer vision, with practical experience from real-world projects at PVCFC. Actively seeking an AI Engineer role to contribute to building, optimizing, and deploying AI solutions in production environments.

EXPERIENCE

PetroVietnam Camau Fertilizer Joint Stock Company (PVCFC) | AI/IT InternJun 2025 – Aug 2025

- Engineered a "Legal Agent" chatbot using Microsoft Copilot Studio to automate responses to common legal inquiries, significantly reducing query resolution time for employees.
- Optimized critical data extraction by analyzing and rewriting complex SQL queries on Risk-Based Inspection (RBI) plant data, enabling faster and more efficient access to operational insights.
- Contributed to the company's data strategy by analyzing the existing ERP system and delivering a comparative report on Data Warehouse vs. Data Lake architectures to inform future data-driven initiatives.

SKILLS

Programming Languages: Python (for Data Science, ML)

ML/DL Frameworks & Libraries: TensorFlow, Keras, Scikit-learn, Pandas, NumPy, OpenCV, Matplotlib

Data & Visualization Tools: SQL Server, Microsoft Excel, Data Cleaning & Visualization

Web Development & Deployment: Flask, FastAPI, Next.js, TypeScript, Vercel, CI/CD & DevOps basics

Platforms / IDEs: PyCharm, Visual Studio Code

Soft Skills: Problem solving, Communication & Presentation, Project management (Agile), Teamwork & Collaboration, Critical thinking, Planning & Prioritization, Self-learning / Adaptability, Stakeholder communication

PROJECTS

Exam Grading System — Automated multiple-choice exam grading; web app & OCRGithub | Nov 2025

Built an end-to-end web system to auto-grade multiple-choice answer sheets: image preprocessing, OCR + vision-based answer detection, review UI, and exportable reports. Frontend: Next.js (TypeScript), Tailwind, shadcn/ui, Zustand. Backend & ML: FastAPI (Python 3.10+), OpenCV, Pandas, Ollama + qwen2.5vl:3b (Vision-Language models). Reduced teacher grading overhead and enabled batch processing + manual correction workflow.

AI Culinary Companion — Intelligent Cooking Assistant powered by Gemini AIGithub | Aug 2025

Developed a modern cooking assistant web app for exploring Vietnamese recipes and generating personalized dishes using Google Gemini AI. Implemented smart search, multi-criteria filtering (ingredients, difficulty, cook time, diet), and responsive recipe browsing with full details and images. Built with React (TypeScript), Vite, TailwindCSS, and integrated Gemini 2.5-flash via the Generative AI SDK to produce structured, AI-generated recipes. Delivered a smooth, mobile-friendly experience for fast recipe discovery and custom meal creation.

Time Series / Oil-Price & Demand Forecasting (LSTM) — Production ML with Unit Testing + CI/CDGithub | May 2025

Developed end-to-end LSTM forecasting system for crude oil price prediction with FastAPI REST API. Implemented comprehensive preprocessing pipeline with 60-day rolling window validation and feature engineering. Built production infrastructure with 95%+ test coverage (pytest), automated CI/CD (GitHub Actions: multi-Python testing, linting, security scanning), and interactive API documentation. Deployed scalable microservice supporting real-time predictions and on-demand model retraining. Tech: Python, TensorFlow/Keras, FastAPI, pytest, GitHub Actions, Pandas, NumPy, scikit-learn.

Ca Mau Fertilizer Classification - Real-time product detectionGithub | Sep 2023

Developed a real-time detection and multi-object tracking pipeline for Ca Mau fertilizer packages using YOLOv8 + DeepSort. Collected/annotated a custom dataset for warehouse/retail scenarios and integrated the model into a live stream processing flow — achieved high classification accuracy in practice (~92% accuracy). Tech: Python, YOLOv8, DeepSort, PyTorch, OpenCV.

EDUCATION

Can Tho University of TechnologyCan Tho, VietNamAug 2021 – Dec 2025

Computer Science; GPA: 3.49 / 4.0

CERTIFICATE & AWARD

Consolation Prize – “Digitalization and Automation Solutions in Agro-Industry” Competition (Q4 2023, Can Tho University)

- [\[Overview\]](#)
- Task: Ca Mau Fertilizer product recognition
- Team size: 5 members

Google AI Essentials – Google (Aug 2025)

- Skills: Artificial Intelligence & Machine Learning (AI/ML), Generative AI, Prompt Engineering, Emerging Technologies, Data Security, Forecasting, Business Process Automation, Organizational Strategy
- [\[View Certificate\]](#)

Google Prompting Essentials – Google (Aug 2025)

- Skills: Prompt Engineering, Generative AI, Spreadsheet Software, Document Management, AI Personalization, Business Correspondence, Report Writing, Data Presentation, Critical Thinking
- [\[View Certificate\]](#)

Foundations of Project Management – Google (Apr 2024)

- Skills: Project Management, Software Project Management, Strategic Thinking, Leadership & Management, Waterfall Methodology, Change Management, Communications Planning, Project Scoping
- [\[View Certificate\]](#)

Scholarship – Can Tho University of Technology

- Awarded for academic excellence
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